

**The Impact of Remote Working on Cybersecurity
Risks and Defence Strategies in Fintech Organisations:
A Qualitative Document Analysis of SOC Effectiveness**

**Module: Project
Module Code: UA92-398**

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Abstract

This dissertation investigates how remote working affects cybersecurity risks, defence strategies and Security Operations Centre (SOC) effectiveness in financial technology (fintech) organisations. As fintech services rely on digital platforms, Application Programming Interfaces (APIs), cloud systems and sensitive financial data, remote and hybrid working can weaken security boundaries and increase exposure to threats. The study uses qualitative document analysis and reflexive thematic analysis to examine academic, industry and regulatory literature on remote-working risks and SOC operations. Six themes were identified: expanded attack surface, Bring Your Own Device (BYOD) and endpoint gaps, phishing and human factors, defence theory versus practice, SOC visibility and alert fatigue, and the under-represented SOC perspective. The findings show that remote working intensifies fintech risks by changing how systems are accessed, monitored and secured. Controls such as Multi-Factor Authentication (MFA), Virtual Private Networks (VPNs), Zero Trust, Security Information and Event Management (SIEM) and endpoint monitoring are necessary, but effectiveness depends on implementation quality, governance, analyst capability and SOC workflow integration. The dissertation argues that SOC effectiveness is the missing link between remote-working risk and defence strategy. It recommends evaluating controls through visibility, alert accuracy, Mean Time To Detect (MTTD), Mean Time To Respond (MTTR), incident response quality and resilience.

Keywords: Cybersecurity, SOC, Defence Strategies, Remote Working, FinTech

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1. Introduction

Financial technology (Fintech) has become central to modern finance, with fintech organisations delivering services through digital banking, mobile payments, peer-to-peer lending, robo-advisory platforms, blockchain and online payment infrastructures. Ali et al. (2024) explain that FinTech uses technologies such as Artificial Intelligence (AI), big data analytics, mobile systems, cloud computing and Application Programming Interfaces (APIs) to deliver financial services more efficiently. However, this digital-first model also increases exposure to cybersecurity threats. Onwuegbuchi et al. (2023) argue that fintech innovation has introduced risks including data breaches, fraud, identity theft, ransomware and API vulnerabilities. Similarly, Pittala and Kacheru (2025) state that fintech firms are attractive targets because they handle sensitive financial information and depend on third-party vendors, open banking APIs and digital platforms.

Cybersecurity is therefore fundamental to trust, compliance and operational continuity in fintech. Kamuangu (2024) identifies data breaches, phishing, malware, regulatory compliance and Multi-Factor Authentication (MFA) as major fintech cybersecurity concerns. Cele and Kwenda (2025) also show that cybersecurity threats such as identity theft, malware, phishing and vishing affect digital banking adoption because users may lose confidence in online financial platforms. These studies suggest that cybersecurity risk in fintech is not only a technical issue, but also a business, regulatory and customer-trust issue.

Remote and hybrid working have further transformed this risk environment. Odeh and Odeh (2025) argue that remote financial workspaces face persistent threats such as phishing, ransomware and credential theft, while basic measures such as MFA and Virtual Private Networks (VPNs) are common but advanced tools such as Zero Trust are still unevenly adopted. This is important because fintech employees may access customer data, payment systems, cloud services and internal applications through home networks and personal devices. As a result, traditional office-based security boundaries become weaker, and cybersecurity teams must manage distributed users, devices and networks.

In response, fintech organisations deploy layered defence strategies including MFA, VPNs, Zero Trust Architecture (ZTA), Security Information and Event Management (SIEM) platforms, endpoint monitoring, encryption, artificial intelligence-based fraud detection and cybersecurity awareness training. Ali et al. (2024) highlight authentication, access control, cryptography, intrusion detection, cloud security, continuous monitoring and regulatory compliance as important mitigation measures. Pittala and Kacheru (2025) also recommend proactive approaches such as Zero Trust, AI-based threat detection, strong encryption, ongoing risk assessment and workforce training. However, the existence of these controls does not automatically prove that they are effective in practice, especially when tools are poorly integrated with monitoring, triage and incident response processes within busy Security Operation Centre (SOC) environments in practice.

This is where SOC effectiveness becomes important. Vielberth et al. (2020) describe SOC as combining people, processes, technologies, governance and compliance to identify, detect and mitigate threats, but they also note that SOC research is fragmented and lacks a fully agreed definition. In fintech organisations, SOC teams carry responsibility because they protect platforms that process real-time payments and sensitive customer financial data. Remote working means SOC teams must detect threats from distributed endpoints, home networks and personal devices, while sometimes working as distributed teams themselves.

1.1 Research Aim

This dissertation uses qualitative document analysis to examine how remote working affects cybersecurity risks, defence strategies and SOC effectiveness in fintech organisations. The aim of this research is to critically investigate how remote working affects cybersecurity risks and defence strategies in fintech organisations, with a specific focus on how these strategies support SOC effectiveness.

1.2 Research Objectives

The objectives of the research are:

1. To define cybersecurity risk, SOC effectiveness and key defence strategies in the context of fintech organisations.

2. To analyse the main cybersecurity risks associated with remote and hybrid working.
3. To evaluate defence strategies such as MFA, VPNs, Zero Trust, SIEM and endpoint monitoring.
4. To identify gaps in the literature around SOC visibility, alert triage and incident response.
5. To recommend improvements for evidence-based SOC resilience in remote fintech environments.

The dissertation reviews literature, explains the methodology, presents themes, discusses implications, and concludes with recommendations for future research and practice.

1.3 Research Question

1. The research question is: How does existing academic, industry and regulatory literature explain the impact of remote working on cybersecurity risks, defence strategies and SOC effectiveness in fintech organisations?

2. Literature Review

2.1 Defining Cybersecurity Risk

Cybersecurity risk refers to the possibility that threats exploit weaknesses in information systems, causing financial loss, service disruption, data exposure or reputational damage. Florackis et al. (2023) define cybersecurity risk as the risk of financial loss, disruption or reputational damage caused by failures in IT systems due to external attacks. This is useful because it links technical incidents to organisational consequences. Putra and Soewito (2023) also show that information security risk management involves identifying, assessing and treating risks through structured standards such as International Organisation for Standardisation (ISO) 27005 and National Institute of Standards and Technology (NIST) Special Publication (SP) 800-30.

In fintech, cybersecurity risk is particularly significant because digital platforms process sensitive financial data and high-value transactions. Kamuangu (2024) identifies data breaches, phishing, malware, authentication and regulatory compliance as major fintech concerns. However, risk is not reduced simply by naming controls. Pomerleau and Maimon (2022) argue that cybersecurity practice should be evidence-based, while Woods and Seymour (2024) show that control effectiveness depends on implementation quality rather than deployment alone.

Risk management frameworks (RMF) adoption also depends on organisational use. Morris (2022) shows that facilitating conditions influence RMF use in cloud systems, while Stoltz (2024) highlights the importance of continuous monitoring within NIST RMF compliance. Therefore, cybersecurity risk should be understood through threats, vulnerabilities, impacts, controls and evidence of practical effectiveness, especially in remote and cloud-based financial environments today.

2.2 Defining SOC and SOC Effectiveness

A Security Operations Centre (SOC) is an organisational function that brings together analysts, processes and security technologies to monitor events, detect threats, investigate alerts and coordinate incident response. Raimondi et al. (2022) show that

SOC capability depends on analyst knowledge, skills and training, while Jones et al. (2023) explain that SOC play a central role in managing networks, defending against cyber threats and maintaining compliance. This suggests that SOC effectiveness cannot be judged by tool deployment alone, but by how well people, processes and technologies work together.

SOC effectiveness can be understood through visibility, alert triage, investigation quality and response speed. Jones et al. (2023) show that work-from-home shifted SOC attention towards endpoint security management, creating additional monitoring, privacy and burnout concerns. Tariq et al. (2025) further show that alert fatigue can reduce analyst effectiveness because SOC teams must manage high volumes of alerts and false positives. Sohal (2025) also links SOC maturity to incident response outcomes, using mean time to detect (MTTD) and mean time to respond (MTTR) to argue that mature, threat-informed SOC can detect and respond faster.

Automation and intelligence-led approaches also shape SOC effectiveness, but these should be evaluated critically. Yousaf and Boomsma (2024) and Martens (2026) suggest that AI-driven tools and real-time threat intelligence may improve threat analysis and situational awareness, but these sources are better used as supporting evidence because they are less established than peer-reviewed journal and conference literature. Similarly, Kearney et al. (2023) provides useful, applied insight into AI-assisted alert fatigue reduction but should be used alongside stronger academic evidence. Therefore, SOC effectiveness depends on maturity, visibility, alert management, skilled analysts, automation quality and practical workflow design.

2.3 Remote Working Cybersecurity Risks across Organisational Context

Remote working has introduced cybersecurity risks across organisational contexts, creating a foundation for understanding fintech risk. Odeh and Odeh (2025) show that remote financial workspaces face phishing, ransomware and credential theft, especially where MFA and VPNs are used but Zero Trust remains unevenly adopted. Atencio (2025) similarly argues that remote work and the Internet of Things (IoT)

expand the attack surface because home networks, cloud tools and connected devices create new entry points.

Bring Your Own Device (BYOD) is another significant risk. Chigada and Daniels (2021) found that personal devices in financial services can expose confidential information when policies, access control and monitoring are weak. Hijji and Alam (2022) also show that remote workers need structured cybersecurity awareness and training, while Angafor, Yevseyeva and Maglaras (2024) demonstrate that regular awareness campaigns can reduce phishing-related risk.

Remote-working risk is therefore both technical and behavioural. Vitla (2024) shows that unsecured Remote Desktop Protocol (RDP) access can enable ransomware, data breaches and corporate extortion, highlighting the danger of poorly secured remote access. Rah (2023) provides useful supporting evidence on BYOD controls such as VPNs, encryption, authentication, antivirus, mobile device management and patching, although it should be treated carefully because it is a doctoral dissertation. Vaka (2020) also supports the link between COVID-19 remote work and phishing, but it should be used cautiously because the publication outlet appears less established than stronger peer-reviewed sources.

2.4 Remote Work in the Fintech Context

The risks documented across general organisations become more consequential in fintech because of sensitive data, payments, third-party platforms and regulatory pressure. Ogunleye et al. (2024) identify data breaches, phishing, malware, insider threats and compliance challenges as major fintech cybersecurity concerns, showing that innovation expands the attack surface. Atencio (2025) adds that remote work, cloud tools and IoT weaken traditional perimeter security, making dynamic and intelligence-led controls necessary.

Financial-sector studies develop this point further. McCarthy (2024) argues that modern finance faces rising cyber risks linked to digitalisation, ransomware, third-party dependence and remote working, while Triesdhany (2026) similarly connects remote work and service digitalisation to data security challenges. However, Triesdhany

should be used carefully because it appears to be an academic paper rather than a peer-reviewed journal article.

Regulation makes these risks more serious. Clausmeier (2023) explains that Digital Operational Resilience Act (DORA) strengthens financial-sector resilience through information and communication technology (ICT) risk management, incident reporting, penetration testing and third-party oversight. The DORA-focused study by Shah (2025) supports this by linking banking-sector exposure to higher ICT incident counts, although it should be treated as supporting evidence because it appears preprint-style. Prakash (2026) also highlights UK financial-service maturity and regulatory pressures but should be used cautiously as a preprint. Overall, remote fintech risk is technical, operational and regulatory rather than simply a remote working issue alone.

2.5 Defence Strategies for Remote Working

Fintech organisations use layered defence strategies to manage remote-working risks, but the literature shows that controls are only effective when implemented and monitored properly. Singh (2025) links MFA, encryption, secure APIs, role-based access control and anomaly detection to transactional integrity in fintech and core banking, showing why defence must protect both access and transaction flows. However, Singh should be used carefully as a book chapter rather than a high-impact journal article.

VPNs remain important for remote access, but they also create risk. Khantamonthon, Patpituck and Chimmanee (2025) show that VPN-enabled environments can still be exposed to ransomware and human error. Zohaib et al. (2024) argue that hybrid work requires Zero Trust VPN approaches because traditional VPNs face security, latency and scalability limits.

Zero Trust provides a stronger model for remote security. Teerakanok, Uehara and Inomata (2021) explain that Zero Trust removes implicit trust and requires continuous verification, although migration can be complex. Tsai, Lee and Shieh (2024) similarly argue that ZTA supports precise authentication, minimal authorisation and continuous monitoring. Gupta et al. (2022) offer useful implementation guidance using NIST controls but should be treated as supporting evidence. Finally, Khan et al. (2024) show

that SIEM prioritisation supports SOC analysts by improving alert triage, response timing and operational decision-making within remote and hybrid fintech environments specifically.

2.6 Summary of Key Literature Reviewed

Table 1 summarises the key literature used in this dissertation and shows how each source contributes to the research problem. It acts as a synthesis tool, helping compare sources by focus, method, findings, limitations and relevance before the written discussion (Fok et al., 2025). The included sources were selected because they directly support the dissertation's core areas: cybersecurity risk, remote working, fintech cybersecurity, defence strategies and SOC effectiveness. Sources not included were either less central, repeated points already covered by stronger evidence, or were used only as supporting context. Some grey literature, dissertations and preprints were therefore not prioritised because peer-reviewed journal articles, conference papers and academic reviews provide stronger evidence. Overall, the table highlights fragmented literature and supports the need to connect these areas through qualitative document analysis.

Table 1: Summary of Key Literature Reviewed

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Florackis et al. (2023)	Cybersecurity risk	Quantitative / financial risk study	Defines cybersecurity risk through financial loss, disruption and reputational damage	Broad cyber-risk focus, not fintech remote work specifically	Supports Section 2.1 by linking technical incidents to organisational consequences

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Putra and Soewito (2023)	Information security risk management	Framework-based study using ISO 27005 and NIST SP 800-30	Shows risk management involves identifying, assessing and treating risks through structured standards	Focuses on insurance sector, not fintech SOC's	Supports the definition of cybersecurity risk and risk management frameworks
Kamuangu (2024)	Fintech cybersecurity risks	Literature review	Identifies data breaches, phishing, malware, authentication and compliance as major fintech concerns	Broad fintech focus with limited SOC-specific analysis	Establishes the fintech cybersecurity risk landscape
Pomerleau and Maimon (2022)	Evidence-based cybersecurity	Conceptual / research-practice discussion	Argues cybersecurity practice should be based on evidence, not assumptions	Broad cybersecurity focus, not fintech-specific	Supports the argument that controls should be evaluated by effectiveness

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Woods and Seymour (2024)	Security control effectiveness	Meta-review	Finds that control effectiveness depends on implementation quality, not deployment alone	Not focused specifically on fintech or remote work	Strengthens the argument that MFA, VPN, Zero Trust and SIEM must be evaluated in practice
Raimondi et al. (2022)	SOC team capability	Conference paper / SOC training focus	Shows SOC capability depends on analyst knowledge, skills and training	Maritime SOC focus, not fintech-specific	Supports Section 2.2 by showing that SOC effectiveness depends on skilled analysts
Jones et al. (2023)	SOC work-from-home challenges	Empirical / qualitative study	Shows work-from-home changed endpoint monitoring, privacy and analyst workload	Not fintech-specific	Supports the SOC effectiveness gap in remote working contexts
Tariq et al. (2025)	Alert fatigue in SOC's	Research review	Shows alert fatigue reduces analyst effectiveness due to high	Broad SOC focus, not fintech-specific	Supports analysis of SOC visibility, alert

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
			alert volumes and false positives		triage and response challenges
Odeh and Odeh (2025)	Remote financial workspaces	Literature and policy-focused review	Identifies phishing, ransomware and credential theft as remote financial-workspace risks	Focuses on financial sector broadly, not fintech specifically	Helps connect remote-working risk to financial services and fintech environments
Chigada and Daniels (2021)	BYOD in financial services	Qualitative study	Finds weak BYOD policies, access control and monitoring can expose confidential information	Small sample and South Africa-specific	Supports endpoint and personal device risk analysis
Hijji and Alam (2022)	Cybersecurity awareness for remote workers	Framework proposal	Shows remote workers need structured cybersecurity awareness and training	General remote-working focus, not fintech-specific	Supports the human-factor side of remote-working risk

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Angafor, Yevseyeva and Maglaras (2024)	Security awareness campaigns	Empirical / awareness campaign study	Finds regular awareness campaigns can reduce phishing-related risk	Focuses on awareness rather than SOC operations	Supports the argument that remote-working risk is behavioural as well as technical
Ogunleye et al. (2024)	Fintech cybersecurity concerns	Systematic review	Identifies data breaches, phishing, malware, insider threats and regulatory concerns in fintech	Limited focus on SOC effectiveness	Supports Section 2.4 by linking fintech innovation to an expanded attack surface
Clausmeier (2023)	DORA and financial-sector resilience	Legal / regulatory analysis	Explains DORA requirements around ICT risk management, incident reporting, testing and third-party oversight	Regulatory focus, not operational SOC analysis	Supports the argument that remote fintech risk is also a compliance issue

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Zohaib et al. (2024)	Zero Trust VPN for hybrid and remote work	Systematic literature review / framework	Finds traditional VPNs face security, latency and scalability limitations in hybrid work	Framework-focused, not fintech SOC-specific	Supports Section 2.5 on VPN limitations and Zero Trust alternatives
Teerakanok, Uehara and Inomata (2021)	Zero Trust Architecture	Review article	Explains that Zero Trust removes implicit trust and requires continuous verification	Notes migration complexity and legacy system challenges	Supports the argument that Zero Trust is useful but difficult to implement
Tsai, Lee and Shieh (2024)	ZTA implementation strategy	IEEE article	Shows ZTA supports precise authentication, minimal authorisation and continuous monitoring	General organisational focus, not fintech-specific	Supports the evaluation of Zero Trust as a remote-working defence strategy
Khan et al. (2024)	SIEM alert prioritisation	Machine learning approach	Shows SIEM prioritisation can support SOC analysts through probability, impact, MTTD and MTTR	Not tested in remote fintech SOC environments	Supports the link between defence tools and SOC alert triage

Author(s)	Focus of Study	Method / Approach	Key Findings	Limitations / Gaps	Relevance to this Dissertation
Singh (2025)	Fintech and core banking transaction security	Book chapter	Links MFA, encryption, secure APIs, role-based access control and anomaly detection to transactional integrity	Book chapter, so use as supporting evidence rather than main evidence	Supports the fintech-specific discussion of layered defence strategies

2.7 Theoretical and Conceptual Framework

This dissertation is guided by socio-technical systems theory, defence-in-depth, evidence-based cybersecurity and SOC maturity. Socio-technical systems theory is the main lens because it explains cybersecurity as an interaction between people, policies, daily tasks and technical infrastructure. Ta, Hoang and Hoang (2025) show that policies, organisational trust and integrating cybersecurity into daily work can improve cybersecurity performance, meaning controls cannot be evaluated separately from organisational behaviour.

Defence-in-depth supports this by framing security as multiple overlapping layers. Jadidi et al. (2023) argue that resilience requires prevention, detection and incident response, while Zero Trust strengthens this by removing implicit trust. This connects to remote fintech environments where MFA, VPNs, endpoint monitoring, SIEM and awareness must operate together.

Evidence-based cybersecurity is also important. Siponen (2025) warns that cybersecurity practices labelled as “best practice” may not always be evidence-based. Therefore, this dissertation evaluates whether defence strategies support SOC effectiveness rather than assuming they work.

Finally, SOC maturity provides an operational framework. Abdel Wahed et al. (2026) and Lubis et al. (2026) show that SOC effectiveness can be assessed through detection coverage, maturity indicators and risk-management frameworks. These concepts guide the analysis of remote-working defence strategies in fintech and support recommendations for more resilient practice.

2.8 Gap in the Literature

The literature review identifies three connected gaps. First, existing studies often examine cybersecurity risk, remote-working threats, defence strategies and SOC effectiveness separately. Fintech risk research highlights phishing, data breaches, compliance and digital-platform exposure (Kamuangu, 2024; Ogunleye et al., 2024), while remote-working studies focus on BYOD, home networks, awareness and remote access risks (Chigada and Daniels, 2021; Angafor, Yevseyeva and Maglaras, 2024).

However, these studies rarely explain how those risks affect SOC visibility, alert triage and incident response.

Second, defence strategies are usually discussed as individual controls rather than as an integrated operational system. Research on MFA, VPNs, Zero Trust and SIEM shows that effectiveness depends on implementation, monitoring and context (Woods and Seymour, 2024; Zohaib et al., 2024; Khan et al., 2024). This leaves uncertainty about whether deployed controls improve SOC effectiveness.

Third, the fintech-specific SOC perspective remains under-developed. SOC studies discuss analyst skills, alert fatigue, maturity and remote work challenges (Jones et al., 2023; Tariq et al., 2025; Lubis et al., 2026), but they are rarely focused on fintech. This dissertation addresses the gap by connecting remote-working risk, defence strategies and SOC effectiveness through qualitative document analysis within remote and hybrid fintech organisational environments and regulatory contexts.

3. Methodology

3.1 Research Approach

This dissertation adopts a qualitative document analysis approach to examine academic and industry sources on remote working, fintech cybersecurity, defence strategies and SOC effectiveness. Bowen (2009) defines document analysis as a systematic process of reviewing and evaluating printed or electronic documents, involving selection, appraisal and synthesis of documentary data. This is suitable because the study investigates how existing evidence explains a cybersecurity problem, rather than collecting primary data from participants. Morgan (2022) argues that qualitative document analysis is useful when researchers analyse pre-existing texts, reduce ethical concerns, and work within practical limits such as restricted access to field sites or participants. This is relevant because fintech SOC operations may involve confidential processes, sensitive information and non-public workflows. A document-based approach allowed the study to examine the topic safely while comparing academic and professional evidence. The approach also supports critical analysis because it enables patterns and gaps across source types to be identified.

3.2 Data Sources and Collection

Documents were collected from academic, industry and regulatory sources. Academic sources were identified through Google Scholar, OneSearch, IEEE Xplore, ACM and USENIX using combinations of “remote working”, “fintech”, “cybersecurity risk”, “SOC effectiveness”, “SIEM”, “Zero Trust”, “VPN”, “BYOD”, “alert fatigue” and “incident response”. The search approach was informed by Sample, Phenomenon of Interest, Design, Evaluation and Research type (SPIDER)-style qualitative searching, which Desty (2025) and Johnson et al. (2022) use to structure eligibility and relevance in qualitative evidence reviews. Industry and regulatory documents were included to capture practice and compliance perspectives where academic literature did not fully address fintech SOC operations. The search strategy documentation is provided in Appendix A. A source-selection log was maintained to record the database or source used, search terms, source type, relevance to the research question and whether the document was used as main or supporting evidence. This made the collection process

transparent and reduced the risk of selecting sources only because they supported one argument.

3.3 Inclusion and Exclusion Criteria

Sources were included if they addressed cybersecurity risk, remote or hybrid work, fintech or financial services, defence strategies, SOC monitoring, alert triage or incident response. Recent sources from 2021 onwards were prioritised for technical topics because remote-working security and SOC practice change quickly. Older sources were included where they were methodologically important or conceptually relevant, such as established work on document analysis. Sources were excluded if they were unrelated to cybersecurity, unsupported, overly general, inaccessible, or lacked enough credibility to support the dissertation argument. Where several sources made similar points, peer-reviewed journal articles, conference papers and recent systematic reviews were prioritised over dissertations, preprints, grey literature or vendor-style sources. Secondary data were used but checked carefully because existing datasets and documents require validity assessment (Seibold, 2024). The search terms and databases used to identify relevant sources are summarised in Appendix A.

3.4 Data Analysis

The documents were analysed through qualitative document analysis and reflexive thematic analysis. Moilanen et al. (2022) recommend identifying purpose, selecting documents, using an extraction matrix, piloting, analysing data, and considering credibility and ethics. Izadi et al. (2023) also demonstrate how policy documents can be selected, appraised and coded into themes. In this dissertation, documents were read repeatedly to identify extracts on remote-working risks, fintech-specific exposure, defence strategies and SOC effectiveness. Initial codes included attack surface, BYOD, phishing, VPN limitations, Zero Trust, SIEM, alert fatigue, analyst workload and regulatory pressure. These codes were then grouped into broader themes that became the basis of Chapter 4. Coding was guided by qualitative coding principles discussed by Lungu (2022), while Braun and Clarke (2021) supported a reflexive

thematic approach focused on coherent and interpretive theme development. To maintain an audit trail, key extracts were organised into a document matrix showing the author, source type, key finding, limitation, relevance and emerging theme. Themes were refined by comparing repeated patterns across sources rather than relying on single-source claims.

3.5 Ethical Considerations

No human participants were recruited, and no confidential organisational data were accessed. These reduced risks linked to consent, privacy and data protection. However, cybersecurity research can still create ethical issues because technical information may be misused. Botes and Lenzini (2022) warn that ransomware research raises dual-use concerns when security knowledge could support harmful activity. Therefore, this dissertation avoids exploit instructions, operational attack detail and confidential SOC information. It also treats vendor claims and weaker publication types critically, especially where sources discuss AI, SIEM or Zero Trust benefits without independent evaluation. Referencing was completed using Harvard style, and AI use is declared in Appendix B.

3.6 Limitations

Document analysis cannot prove causation or capture unpublished SOC practice. It also cannot confirm how individual fintech SOC teams experience remote working. Industry sources may contain bias, and available academic evidence is uneven across fintech, remote work and SOC effectiveness. To reduce this weakness, academic, industry and regulatory documents were compared transparently, and source limitations were recorded during analysis. This maintained transparency, but the findings should be understood as an evidence-based interpretation of existing documents rather than primary organisational measurement.

4. Findings and Analysis

Thematic analysis identified six connected themes across the selected documents. These themes show how remote working reshapes fintech cybersecurity risk and why SOC effectiveness is central to evaluating defence strategies.

4.1 Theme 1: The Expanded Attack Surface in Remote Work

Remote work expands the attack surface by moving organisational activity beyond managed offices into home networks, cloud platforms and dispersed endpoints. Ogunleye et al. (2024) show that fintech innovation already exposes organisations to data breaches, phishing, malware and insider threats, while Singh (2024) focuses more directly on endpoints as attack vectors in distributed work. Ferreira et al. (2025) adds an organisational perspective by showing that financial institutions require coordinated assessment of cybersecurity controls. The comparison between these sources suggests that remote fintech risk is not simply caused by working from home. Instead, it emerges when digital financial services, remote access, third-party systems and weak control assurance overlap. This matters because fintech environments depend on payment workflows, customer data, APIs and identity systems that require continuous monitoring. Therefore, attack surface expansion should be understood as both a technical and governance issue.

4.2 Theme 2: BYOD and Endpoint Security Gaps

BYOD and endpoint gaps were also prominent. Ngcai and Gundu (2025) present BYOD as a source of flexibility in financial organisations, but also identify challenges around device diversity, sensitive data and compliance. Njuguna and Kanyi (2023) provide a more practical warning, finding low awareness and neglected BYOD safeguards in a case study. Singh (2024) offers technical mitigation through endpoint detection, MFA, VPNs and Zero Trust. Comparing these sources shows that endpoint risk is socio-technical: personal devices create exposure, but weak policies, poor awareness and limited SOC visibility make that exposure harder to control. In fintech, this is especially important because unmanaged or poorly monitored endpoints may

connect to high-value financial systems. The finding therefore supports the argument that endpoint protection must be linked to monitoring, logging and clear user responsibility, rather than treated as a device-management issue only.

4.3 Theme 3: Phishing and Human Factors

Human factors shaped remote-working risk across several sources. Karayel, Aktas and Akbıyık (2025) found that organisational cybersecurity measures are linked to remote workers' cyber hygiene, while Poduval, Shenbhagavadivu and Vinitha (2025) argue that organisational and technological support matter more than work location itself. Begam and Dasgupta (2026) extend this to banking by linking security-related citizenship behaviour with better control effectiveness and anomaly reporting. These studies agree that user behaviour matters, but they move beyond blaming employees. Instead, they show that secure behaviour depends on training, workplace support, culture and usable security processes. This is significant for fintech SOC's because phishing alerts, unusual login activity and policy violations cannot be interpreted only as technical events. They also reflect pressure, awareness, fatigue and organisational context. Therefore, human-factor risks shape alert quality and the meaning of SOC investigations.

4.4 Theme 4: Defence Strategies Between Theory and Practice

A fourth theme was the gap between defence strategies in theory and practice. Khantamonthon, Patpituck and Chimmanee (2025) show that VPN environments can still support ransomware risk where human error and governance weaknesses remain. Ferreira et al. (2025) similarly argue that controls require independent assessment, not just deployment. Hossen (2026) reports stronger banking fraud detection through AI-enhanced SOC systems than traditional SIEM, although this evidence should be treated carefully because the publication outlet appears less established than stronger peer-reviewed sources. Taken together, these findings challenge tool-centred assumptions. VPNs, SIEM, AI and Zero Trust can improve security, but only when they are implemented within reliable workflows, audit processes, monitoring coverage and analyst decision-making. This supports the finding that defence effectiveness is

conditional, not automatic. For fintech organisations, the key issue is whether controls improve real operational outcomes, rather than whether they are present in policy documents.

4.5 Theme 5: SOC Visibility, Alert Fatigue and Incident Response

SOC visibility, alert fatigue and incident response formed the strongest operational theme. Alahmadi, Axon and Martinovic (2022) show that SOC analysts face high volumes of false-positive alarms requiring manual validation, while the SANS survey identifies detection, monitoring, incident response, alert triage and staffing as core SOC challenges (Crowley and Filkins, 2022). Ndichu et al. (2026) and Chowdhury and Tanvir (2026) propose AI-supported triage, but as preprints they should support rather than replace stronger evidence. The contrast is important because industry evidence highlights everyday operational pressure, while academic work explains why alert quality and decision support affect analyst effectiveness. In remote fintech settings, incomplete endpoint telemetry or poorly contextualised alerts may delay response to fraud, credential misuse or payment-system threats. This shows that SOC effectiveness depends on the quality of information reaching analysts and the processes used to prioritise action.

4.6 Theme 6: The Under-Represented SOC Perspective

The final theme was the under-represented SOC perspective. Hyppänen (2024) shows that SOC service adoption can fail when responsibilities and workflows are misunderstood, although this master's thesis is supporting evidence only. Gwavava and Morin (2025) demonstrate that KPI-driven SOC evaluation can misalign analyst behaviour with wider organisational goals. Nasr et al. (2025) show that sector-specific SOC adaptation remains underdeveloped in maritime cybersecurity, while Ferreira, Pacheco and Almeida (2026) propose a maturity-tiered SOC design for resource-constrained organisations, though this is a preprint. Together, these sources confirm that SOC effectiveness is highly contextual. For fintech, this means the key gap is not whether controls exist, but whether they improve visibility, triage and response in remote environments. Overall, the findings confirm that remote-working risks and

defence strategies should be analysed through SOC effectiveness because SOCs connect security controls to practical detection, investigation and response outcomes. This also strengthens the dissertation's central argument that evidence-based cybersecurity requires attention to people, process, technology and context, not the isolated presence of security tools within remote fintech SOC practice today.

5. Discussion

5.1 How Remote Working Reshapes Fintech Cybersecurity Risk

The findings show that remote working has not created completely new cybersecurity risks, but has reshaped existing risks by changing where, how and by whom fintech systems are accessed. Earlier literature identified fintech exposure to data breaches, phishing, malware, insider threats and regulatory pressure, while the findings show that remote work intensifies these issues through home networks, personal devices, cloud services and distributed endpoints (Ogunleye et al., 2024; Singh, 2024). This supports the socio-technical framework because risk is produced through the interaction between technology, user behaviour, organisational support and governance. For fintech organisations, this reshaping is serious because remote access may connect directly to payment systems, customer data, APIs and identity platforms. Therefore, remote working should not be treated as a simple location change, but as a change in the whole cybersecurity risk environment. The discussion also shows that fintech firms must consider how everyday working practices affect security monitoring, because risks increase when employees, devices and systems operate outside traditional organisational boundaries.

5.2 The Conditional Effectiveness of Defence Strategies

The findings also confirm that defence strategies are necessary, but their effectiveness is conditional. Controls such as MFA, VPNs, Zero Trust, SIEM, endpoint monitoring and AI-assisted detection are repeatedly presented as important protections. However, the literature and findings show that deployment alone does not prove effectiveness. VPN environments may still be affected by ransomware and human error, while SIEM and AI tools depend on alert quality, governance and analyst interpretation (Khantamonthon, Patpituck and Chimmanee, 2025; Hossen, 2026). This supports Woods and Seymour's (2024) argument that control effectiveness depends on implementation context. For fintech, this means organisations should not only ask whether a control exists, but whether it improves visibility, reduces false positives, supports investigation and strengthens response. A defence strategy is therefore effective only when it supports real operational outcomes, not just compliance or

procurement evidence. This is particularly important in fintech because controls that appear strong on paper may still fail if they are poorly integrated with SOC workflows, user behaviour, incident reporting and regulatory expectations.

5.3 SOC Effectiveness as a Missing Link

The most important discussion point is that SOC effectiveness is the missing link between remote-working risk and defence strategy. Literature on fintech cybersecurity often explains risks, while literature on security controls explains technical solutions. However, Chapter 4 showed that fewer sources explain how those controls help SOC analysts monitor, triage and respond in practice. This matters because SOC turn technical signals into security decisions. If alerts are incomplete, noisy or misaligned with business context, controls may exist without producing better detection. Alahmadi, Axon and Martinovic (2022) and Crowley and Filkins (2022) show that false positives, staffing and triage remain major SOC challenges. Jones et al. (2023) also show that remote SOC work affects endpoint monitoring, workload and privacy. These findings suggest that fintech SOC effectiveness depends on complete telemetry, clear workflows, analyst capability, realistic KPIs and decision support. It also reflects the gap identified in Chapter 2, where fintech risk, remote work, technical controls and SOC operations were often discussed separately rather than as one connected problem. Therefore, this dissertation argues that SOC effectiveness should be studied as a distinct issue, rather than assumed from control deployment. This strengthens the originality of the project because it connects cybersecurity risk, defence strategy and operational security practice.

5.4 Implications for Practice

The practical implications are clear. First, fintech organisations should evaluate remote-working controls by their contribution to SOC visibility, alert triage, MTTD, MTTR and incident response quality. Second, they should design SOC as socio-technical systems, balancing investment in tools with investment in analyst training, collaboration, wellbeing and governance. Third, BYOD and endpoint security should be treated as monitoring problems as well as access-control problems, because

unmanaged devices can reduce the quality of SOC evidence. Fourth, vendor claims about AI, SIEM or Zero Trust should be tested against independent evidence and organisational context. Finally, fintech firms should align technical controls with regulatory expectations such as resilience, auditability and incident reporting. Overall, the discussion shows that improving cybersecurity in remote fintech environments requires more than adding tools. It requires evidence-based assessment of whether those tools genuinely support SOC detection, triage and response. This makes SOC performance the practical measure of whether remote-working security is genuinely resilient, rather than merely documented as compliant within policy or technology procurement decisions.

6. Conclusion

This dissertation has shown that remote working has reshaped cybersecurity risk in financial technology organisations by moving access to sensitive data, payment systems, cloud platforms and application programming interfaces beyond traditional office boundaries. The central argument is that remote working does not create entirely new cybersecurity risks; instead, it intensifies existing fintech risks and makes them harder to monitor, investigate and control.

Using qualitative document analysis and reflexive thematic analysis, the study identified six connected findings: an expanded attack surface, Bring Your Own Device and endpoint gaps, phishing and human factors, the gap between defence theory and practice, SOC visibility and alert fatigue, and the under-represented SOC perspective. Together, these findings show that cybersecurity risk in remote fintech environments is socio-technical because it depends on people, devices, workflows, governance and security technologies working together.

The main contribution of this dissertation is the argument that Security Operations Centre (SOC) effectiveness is the missing link between remote-working risk and defence strategy. Controls such as multi-factor authentication, virtual private networks, Zero Trust, security information and event management, endpoint monitoring and AI-assisted detection are important, but they only improve security when they support visibility, alert triage, mean time to detect, mean time to respond and incident response quality.

The study is limited because it relies on existing literatures rather than primary data from fintech SOC teams. However, this approach was appropriate because SOC operations can involve confidential and sensitive information (Pittala and Kacheru, 2025; Vielberth et al., 2020). Future research should therefore test these findings through interviews, case studies or surveys with SOC analysts in remote and hybrid fintech settings.

Overall, the lesson from this dissertation is that resilient fintech cybersecurity cannot be achieved by adding tools alone. It requires evidence-based evaluation of whether those tools genuinely help SOC teams detect, investigate and respond to threats.

7. Recommendations and Further Research

Based on the findings, fintech organisations should evaluate cybersecurity controls through operational SOC outcomes rather than deployment statistics alone. This means measuring whether controls improve MTTD, MTTR, alert accuracy, escalation quality, investigation speed and incident response effectiveness. Defence strategies should also be assessed as an integrated system, recognising that MFA, VPNs, Zero Trust, SIEM, endpoint monitoring and awareness training depend on each other. A control may appear effective in isolation but fail to support resilience if it does not provide useful telemetry, reduce false positives or help analysts make faster decisions.

Fintech firms should treat BYOD and endpoint security as monitoring challenges, not only access-control issues. Unmanaged or poorly configured devices can reduce evidence quality and make incidents harder to investigate. Therefore, organisations should strengthen endpoint visibility, device compliance, logging and user accountability. Investment in analyst training, collaboration tools, wellbeing and clear workflows should match investment in security technologies. This is important because SOC effectiveness depends on people and processes as much as tools. Vendor claims about AI, SIEM or Zero Trust should be tested against independent evidence, risk assessments and SOC performance. Fintech organisations should align SOC capabilities with regulatory expectations for resilience, auditability, third-party risk management and incident reporting.

Further research should include empirical studies of SOC operations in remote and hybrid fintech environments. Interviews, case studies or surveys with SOC analysts could provide deeper insight into real monitoring challenges, alert fatigue and remote collaboration. Longitudinal studies could examine how SOC maturity, analyst workload and control effectiveness change over time. Comparative research across fintech organisations of different sizes would also be useful, because smaller fintech firms may lack mature SOC resources. Finally, future research should examine how distributed SOC collaboration, analyst wellbeing and alert fatigue affect security outcomes in real fintech settings, supporting stronger evidence-based resilience in practice.

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Appendices

Appendix A: Search Strategy Documentation

The search strategy was carried out between June and July 2026 using academic, industry and regulatory sources. Searches focused on remote working, fintech cybersecurity, defence strategies and SOC effectiveness. Search terms were refined during the process to include BYOD, phishing, SIEM, Zero Trust, alert fatigue, SOC maturity and operational resilience. Peer-reviewed sources were prioritised, while industry reports and grey literature were used as supporting evidence where relevant.

Database / Source	Search Terms Used	Date / Search Period
Google Scholar	"Remote working" AND "fintech" AND "cybersecurity"	June–July 2026
Google Scholar	"Remote work" AND "cybersecurity risks" AND "financial services"	June–July 2026
Google Scholar	"Security operations centre" AND "remote work"	June–July 2026
Google Scholar	"SOC effectiveness" AND "cybersecurity"	June–July 2026
Google Scholar	"BYOD" AND "financial services" AND "cybersecurity"	June–July 2026
Google Scholar	"phishing" AND "remote work" AND "cybersecurity awareness"	June–July 2026
University Library / OneSearch	"Zero Trust" AND "remote work" AND "financial services"	June–July 2026
University Library / OneSearch	"Document analysis" AND "qualitative research"	June–July 2026

Database / Source	Search Terms Used	Date / Search Period
IEEE Xplore	"SIEM" AND "remote work" AND "financial services"	June–July 2026
IEEE Xplore	"SOC maturity" AND "incident response"	June–July 2026
ACM / USENIX / SANS sources	"SOC alert fatigue" AND "alert triage"	June–July 2026
Regulatory and industry sources	"DORA" AND "financial sector" AND "operational resilience"	June–July 2026

Appendix B: AI Usage Declaration

ChatGPT was used as an academic support tool during the preparation of this dissertation. It was used to support brainstorming, planning, explanation of concepts, report structure, search phrase development, proofreading and refinement of academic writing.

All AI-assisted content was reviewed, edited and adapted by me. The final topic, research question, objectives, argument, source selection, critical evaluation and submitted work reflect my own understanding and academic judgement. AI was not used to fabricate sources, create false references, collect primary data or replace my own analysis.